A User’s Manual

To

TIMCPOT: Tcl/Tk Interface for Measuring Crustal Phase Onset Time

Version R1.0

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Contents

File list

./bin/

./example/data/20110620.171.10.16.49.000_5.3
./example/data/20110809.221.11.50.17.000_5.2
./example/station.loc.txt

./src/evtlocate.f90
./src/fftsub.f90
./src/hilbertsac.f90
./src/libsun.f
./src/libtau.f
./src/makefile
./src/sacio.a
./src/timcpot.tcl
./src/timesub.f90
./src/ttlim.inc

./sup.soft/ttimes.tar.gz

./manual.pdf

Installation

TIMCPOT runs on Linux OS. At present, we have tested it in the following Linux OS:
Red Hat Enterprise Linux Server 6.4
Red Hat Enterprise Linux Server 6.10
Red Hat Enterprise Linux Server 7.6
CentOS 6.6
We didn't conduct more tests on other platforms, but it doesn't mean that TIMCPOT does not work on other platforms. If you have applied it on a platform successfully rather than those as listed above, please let us know, we appreciate your feedbacks.

Requirements of OS platform
OS: Linux
Disk space: <100Mb
CPU: 2.0GHz
RAM: >2GB
Screen: 1920 x 1080

TIMCPOT depends on SAC, ttimes and Tcl/Tk packages. These packages should be installed before running TIMCPOT.

Install SAC

We recommend to build SAC from source code. More details on how to install SAC, please refer to the following websites:
http://www.iris.edu/software/manual.html
https://seisman.github.io/SAC_Docs_zh/install/linux-source/

Install ttimes

A modified version of *ttimes* package can be found in sup.soft folder, named ttimes.tar.gz.
Unzip this package to a directory (e.g., ~/prog/ttimes).
Add the following line to ~/.bashrc or ~/.bash_profile
export HOMETTIMES=~/prog/ttimes
The environment variable, HOMETTIMES, should be correct. It is not only required by ttimes/ttimes.f, but also required by src/evtlocate.f90.
Next, open a terminal and follow the commands to finish installation.

```
source ~/.bash_profile    or    source ~/.bashrc
cd ~/prog/ttimes
make
./remodl iasp91
./setbrn
./remodl ak135
./setbrn
cp ttimes ~/bin/
```

Install Tcl/Tk

TIMCPOT depends on the latest release version of Tcl/Tk packages, named tcl8.6.12-src.tar.gz and tk8.6.12-src.tar.gz. User can download them from https://tcl.tk/software/tcltk/download.html.
Additionally, a copy of Tcl/Tk packages can be found in sup.soft folder. Note, if user obtains TIMCPOT package through supplemental material, user would not find Tcl/Tk packages in sup.soft folder due to a limitation of maximal file size.
tcl8.6.12-src.tar.gz and tk8.6.12-src.tar.gz should be unzipped into an identical folder. Otherwise, user may need to tell Tk where Tcl was built with the *--with-tcl* flag when user configure Tk. See more details at https://tcl.tk/doc/howto/compile.html.
Here is a brief introduction for how to compile Tcl/Tk from source distribution in my Linux platform, and user can take it as a reference.
Unzip Tcl/Tk packages into ~/prog/, and then the directory structures should be listed as:
~/prog/tcl8.6.12
~/prog/tk8.6.12

Next, open a terminal and follow the commands to finish installation.
For Tcl:

```
cd ~/prog/tcl8.6.12/unix
./configure --prefix=/home/xxx/prog/tcl8.6.12
make
make install
```

Note: characters 'xxx' should be replaced by which are actually shown in user's computer.

For Tk:

```
cd ~/prog/tk8.6.12/unix
./configure --prefix=/home/xxx/prog/tk8.6.12
make
make install
```

Note: characters 'xxx' should be replaced by which are actually shown in user's computer.

Add the following two lines to ~/.bashrc or ~/.bash_profile

```
export PATH=~/prog/tcl8.6.12/bin:$PATH
export PATH=~/prog/tk8.6.12/bin:$PATH
```

Install TIMCPOT

Copy TIMCPOT package to a folder (e.g., ~/prog/)
copy sacio.a from SACHOME/lib/ to TIMCPOT/src.
Next, open a terminal and follow the commands to finish installation.

```
cd ~/prog/TIMCPOT/src
make
```

Add the following line to ~/.bashrc or ~/.bash_profile

```
export PATH=~/prog/TIMCPOT/bin:$PATH
```

Demonstration

An example is used to check whether all the requirements are available or workable. In compliance with data protection regulations, station's coordinate stored in example data folder has been rounded to one decimal place. Furthermore, data sample is decimated to an interval of 0.1 s. So, this data is only used for test.

After installation, follow the next commands to check if it is workable.

```
source ~/.bash_profile    or    source ~/.bashrc
cd ~/prog/TIMCPOT/example
mkdir outdir
timcpot.tcl data/ outdir/ station.loc.txt
```

A graphical interface will show on user's computer screen. Once user finishes the following operations (Step1 -Step6) and if user finds an appearance like to Figure 1, it would indicate that all of the requirements are available and workable.

Step1: click at one of the event ID lists

Step2-Step4: click Load button

Step5: press left mouse button and drag mouse cursor

Step6: check all of three checkboxes

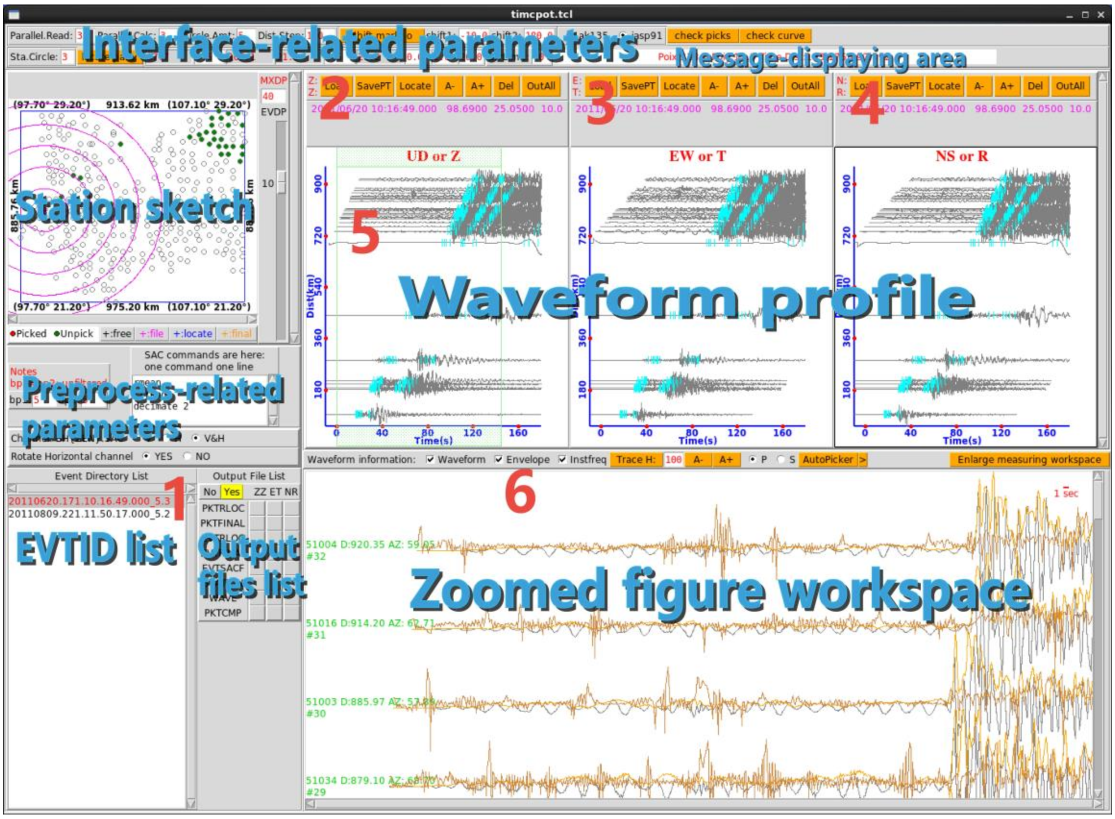
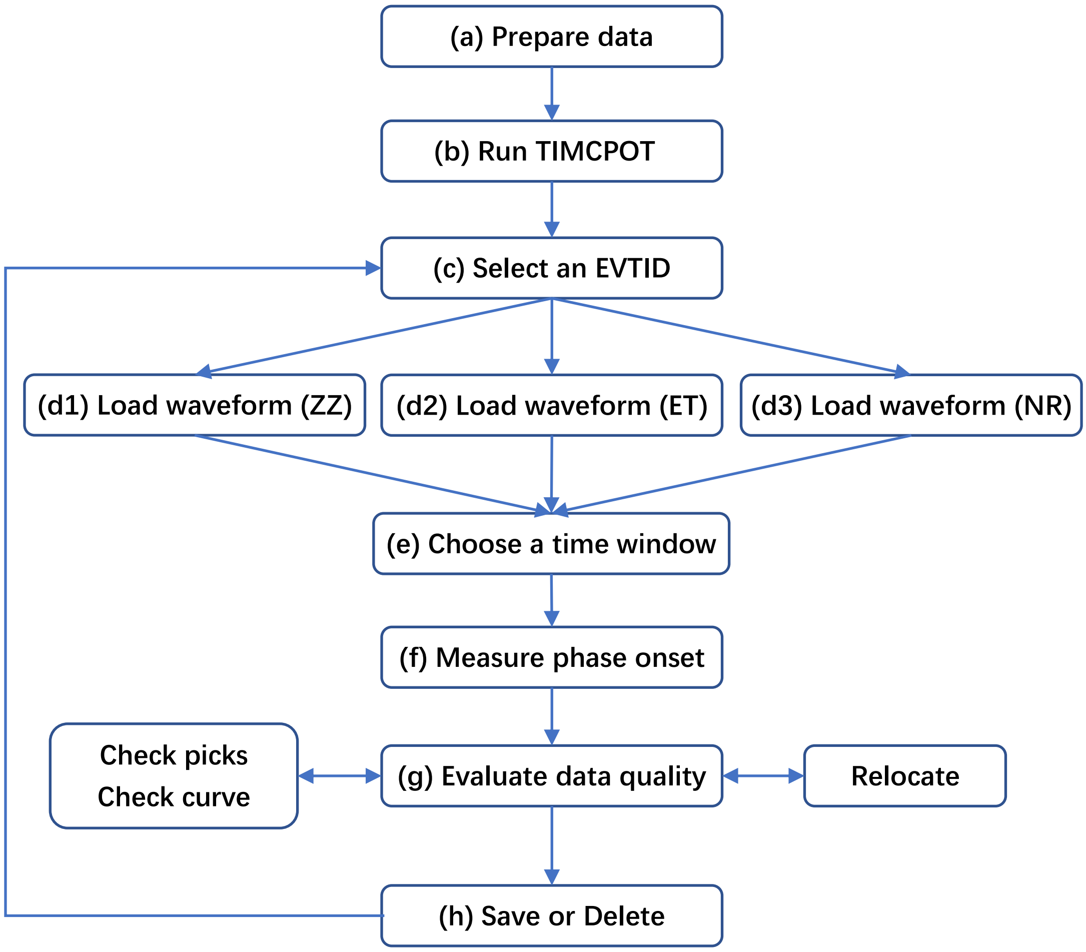


Figure 1. Screenshot of TIMCPOT and explanatory notes

General workflow



A general chart for TIMCPOT workflows is shown as above. In this section, every workflow of TIMCPOT is briefly introduced. User may refer to next several sections for more details, especially for how to set user-defined parameters.

(a) Prepare data. Before running TIMCPOT, user should prepare seismic data according to what are introduced in the next section. Data preparation includes seismic data format, event and station information, and filename format. Two events waveform data has been ready for the demonstration and is stored in the directory "example/data".

(b) Run TIMCPOT and create a graphical interface. TIMCPOT is called from terminal command line that is followed by three parameters (data directory, output directory, and a plain text file for station information). e.g., `timcpot.tcl data/ outdir/ station.loc.txt`. Then a graphical interface will be created and user will see a station sketch map in left top corner of this interface. Each folder name, which is stored in 'data/', is regarded as an EVTID and loaded into a list box (as shown in left bottom corner of the interface).

(c) Select an EVTID for measurement. Through single click of left mouse button, user can select an EVTID. Once an EVTID is selected, the font color of selected EVTID will become red and TIMCPOT will start to preprocess the corresponding seismic data. Meanwhile TIMCPOT's interface will be frozen for a moment until this procedure is completed. After this procedure, the preprocessed data is stored in a folder name swap.

(d) Load waveform. According to different demands, user can load one(s) of three waveform components. After the completeness of load procedure, a waveform profile sorted by epicentral distance will appear in corresponding figure panel. Also, user will notice that some (or all) station signs will be filled with green. Those green-colored stations have waveform data for the selected event while the unfilled stations do not have available data.

(e) Choose a time window and zoom in waveform details. By pressing left mouse button and dragging mouse cursor from left to right, user can choose a time window to zoom in waveform details within this time duration. The zoomed waveforms are shown in right bottom corner of the interface. Note that, the zoomed waveform may not appear only if user selects ☐waveform. The waveforms can be zoomed in/out again in the zoomed figure workspace. To zoom in, press and drag left mouse button, while to zoom out, click right mouse button.

(f) Measure desired phase onset. User can double-click left mouse button to mark a tick for a desired phase onset time on each station waveform. User will notice that the color filled in station signs will become red. Dark-red signs represent those stations which have a relatively large arrival time, while light-red signs represent relatively small arrival time. An AutoPicker is available at this stage. User can utilize AutoPicker button to mark the phase arrival of interest.

(g) Evaluate data quality. The manual picks can be evaluated either by analyzing the consistency between current and previous picks, or by checking the smoothness of traveltime curve, or by both. The corresponding functional buttons are check picks and check curve. If data quality is evaluated to be less well, user can improve it by revising the anomalous picks. If the picked data is confirmed to be without any ambiguities, then user can optionally decide whether to improve their quality again by mean of relocation.

(h) Output/discard measurement results. For a specific event, as the measurement work is completed, user would decide to either save or discard the measurement results. These operations are achieved, respectively, through OutAll button and Del button. If click OutAll button, the corresponding meta information are exported to several plain text files. If click Del button , only the measurement results are discarded, while the event data is reserved as its original being. Actually, TIMCPOT doesn't modify any original seismic data. This design concept is helpful for future repeated checking.

The above steps are the basic workflows for measuring a single event data. User can repeat steps (c) - (h) to deal with more events data.

Data preparation

The data format and necessary parameters

TIMCPOT supports one type of seismic data format at present. It is SAC format, widely used in seismological community. Several kinds of necessary parameters for both events and stations are saved as variables in data header of SAC file.

The following header variables should be assigned with a correct value.

Station information: *stla*, *stlo*, *stel*, *kstnm*

Event information: *evla*, *evlo*, *evdp*, *o*, *mag*

Channel information: *cmpaz*, *cmpinc*

Data information: *delta*, *kzdate*, *kztime*

Note: User can refer SAC manual file for more details of these header variables. The unit for station and event coordinate is degree and the unit for station elevation and event depth is kilometer. The data samples of all SAC files should be unified into an identical interval. All reference times of SAC files associated with an identical event should be unified to an identical reference date and time. For an identical event, if *delta*, *kzdate* and *kztime* are assigned with inconsistent values, it will result in unforeseen mistakes.

Directory structure and filename format

The SAC files are sorted by event. Events should be identified with IDs, name EVTID. Station should also be distinguished by IDs, usually taking station name (*kstnm*) as STAID. The directory structure is organize as a pattern as "data/EVTID/EVTID.STAID.BH?.SAC". EVTID is used as a folder name, the combination, EVTID.STAID.BH?.SAC, is used as filenames of SAC files. An example shown in demonstration is

"data/20110620.171.10.16.49.000_5.3/20110620.171.10.16.49.000_5.3.51002.BHZ.SAC".

(1) "20110620.171.10.16.49.000_5.3" is EVTID. EVTID consists of a time point and event magnitude. e.g., this event occurred on 20 June, 2011, at 10:16:49.000, with magnitude of 5.3. 171 is Julian day.

(2) "51002" is STAID, and is equal to *kstnm*, a header variable stored in SAC file. This part must be also equal to station name that was stored in example/station.loc.txt. Note: STAID field cannot include dot ".", otherwise, unforeseen mistakes will be introduced.

(3) "BHZ.SAC" contains information on seismic data channel and data format. This part is mandatory to be one of "BHZ.SAC", "BHE.SAC" and "BHN.SAC".

Generally, EVTID is not fixed, user can define their preferred format for EVTID. STAID must be equal to station name, and this field is limited to no more than 8 characters, as well, dot "." is forbidden. The third part is fixed to 7 characters, and should be one of "BHZ.SAC", "BHE.SAC", and "BHN.SAC". For more details about directory structure, user can refer to what are stored in example/data.

Station information

User should prepare station information according to what is stored in example/station.loc.txt. There are four columns separated by blank space(s). The text format is not fixed, but station information must be listed in an order as name, latitude, longitude, and elevation.

Procedure and brief introduction

In a terminal command window, change directory to TIMCPOT/example, create a folder with name outdir, input commands

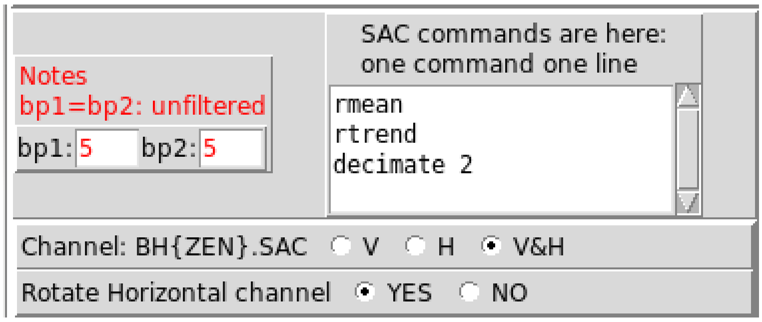
```
source ~/.bash_profile    or    source ~/.bashrc
cd ~/prog/TIMCPOT/example
mkdir outdir
timcpot.tcl data/ outdir/ station.loc.txt
```

A graphical interface will display as Figure 1. The top two lines are for interface-related parameters. Details for these parameters will be explained in the subsequent procedures. Please remember there is a uniform regulation: if a parameter values with red font is modified in any textbox with white background, user should click right mouse button to make it available throughout TIMCPOT. Meanwhile, a message is shown in message-displaying area.

Once the graphical interface is created, a list for event ID will be loaded into TIMCPOT, as what is shown in the left bottom corner of TIMCPOT. A sketch for station position is also shown in the left top corner.

Parameters for preprocessing

Before processing seismic data, user can set up some parameters for preprocessing.



Bandpass filter parameters are set at bp1 and bp2. If bp1 is equal to bp2, seismic data will be unfiltered. Otherwise, data will be filtered by the aid of SAC package following a command as "*bp co bp1 bp2 p 2*". Other SAC commands are also available. User can type SAC commands in the textbox. It should be one command one line.

Sometimes, user just needs to deal with vertical/horizontal channels. To save computation time, three radio buttons are designed to fulfil this demand. V is only for vertical; H is only for horizontal; V&H is for both vertical and horizontal. In addition, user can choose whether or not to rotate horizontal channels via another group of radio buttons.

Processing data

Once user clicks on any event ID, TIMCPOT will follow preprocessing parameters to deal with seismic data. Finally, the prepared data will be saved in a temporary folder, named swap. During this procedure, TIMCPOT will run in parallel mode. The amount of thread used in parallel mode is set at Paralle.Read and Parallel.Calc. Where, the first value is for the amount of thread used to read SAC file; the second value is for the amount of thread used to preprocess seismic data. Due to the I/O restrictions of hard disk, the amount of thread for reading cannot be greater than those for preprocessing. I recommend 3 for both values.

Displaying station and event information in sketch

After clicking Load button, the sketch for station (Figure 2) is updated. Station positions are marked as circles, and those stations with waveform records are filled with green/red color. The circle size can be changed by modifying the value beside Sta.Circle. The big concentric circles denote epicentral distance. The amount of concentric circles and interval between two adjacent circles are given in the values beside Circle.Amt and Dist.Step.

The event position is marked as a cross '+' in station sketch. As you could see, the amount of cross is more than one and these crosses are distinguished by different colors. The black one is manually free, user can move it to a desired position by clicking left mouse button. The magenta one denotes the event location stored in SAC file. The blue one marks location of event derived from relocation procedure. The yellow one represents the final location exported by TIMCPOT. Note that, the relocation procedure embedded in TIMCPOT can determine event location using two algorithms. That means, after relocation procedure, two positions (two blue crosses) will be obtained for that event. If the two position is solved out to be an identical position, they should overlap. Thus, the amount of cross is five at most.

Not all of the crosses are always visible. Firstly, event positions from different sources would be consistent with each other. e.g., it is even possible that relocation procedure could result in a position same to that stored in SAC file. It means that the crosses may overlap at an identical position, and it is often for two blue crosses. Secondly, user should take attentions, the relocation positions (two blue crosses) are not available/visible before conducting relocation procedure, and it is the same case for the final location (yellow cross) before clicking OutAll button. Moreover, event positions may be placed out of the current scope of station sketch. User can check whether event positions are available in a gray-shaded board on top of the waveform profile (Figure 3). The font colors in this board have the same meanings to those for crosses in station sketch. The gray shaded circle denote location uncertainty determined by bootstrap method.

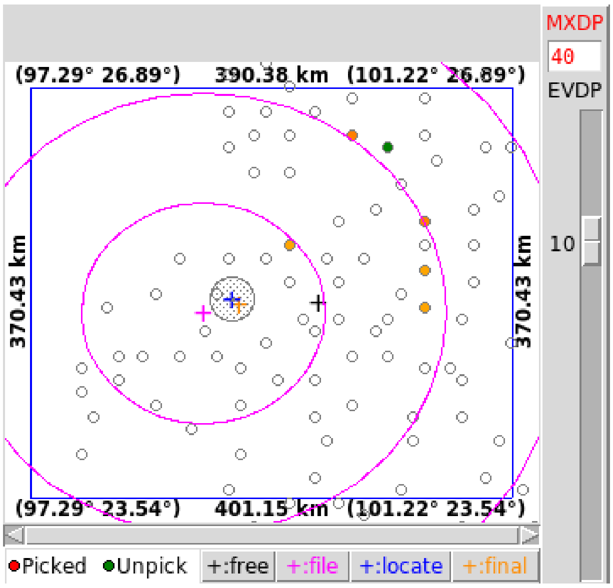


Figure 2. Station sketch panel

A scale to the right of station sketch is used to show event depth corresponding to a hypocentral position that is marked with big concentric circles. The colors of concentric circles are also same to those for crosses in station sketch. Actions and its corresponding operations in station sketch are listed in Table 1.

Table 1. Actions and operations in station sketch

Actions	Operations
Click left mouse button	Move black cross to mouse cursor position
Press & drag left mouse button	Move station sketch
Scroll mouse wheel	Zoom in/out station sketch
Click right mouse button	Turn on measuring distance
Double-click right mouse button	Turn off measuring distance
Click right mouse button on event cross	Move concentric circles to current event cross; Update waveform profile
Press key-x	Disable/enable green lines following mouse cursor
Click middle mouse button on stations	Select/deselect a station to highlight its waveform (if its waveform is available)
Click middle mouse button on concentric circles	Delete concentric circles

Displaying seismic data in waveform profile

After clicking Load button, waveforms are loaded as what is shown in Figure 3. In this profile, waveforms are sorted by epicentral distance and are aligned with a specific time marker. The time marker is chosen from button shift mark: o, the nearby two values, shift1 and shift2, are used to determine the length of waveform time widow. Waveforms are shown as dark-gray curves. Manual picks are shown as red ticks if they are available. Theoretical predictions are shown as cyan ticks. Theoretical predictions are derived from 1D velocity model through *ttimes* package. Optional velocity models are ak135 and iasp91, which can be switched through a group of radio buttons ⊙ak135 ⊙iasp91.

On the top of waveform profile panel, there are a gray-shaded board and a row of buttons. The board is used to shown event information as mentioned above. The functions of these buttons are listed in Table 2. The actions and its corresponding operations in waveform profile are listed in Table 3.

Table 2. Functions of buttons

Button	Function
Load	Load seismic data; Update station sketch; Create waveform profile
SavePT	Save manual picks to output folder as a file with file extension as PICK.FORLOC This file contains manually picking traveltime for relocation procedure
Locate	Save manual picks; Relocate hypocentral position; Output relocation results to output folder as a file with file extension as EVENT.RLOC
A-	Scale down waveform amplitude
A+	Scale up waveform amplitude
Del	Delete all of the output files corresponding to the selected event ID
OutAll	Output results, including: PKTFINAL: manual picks, meta information for tomographic study

	EVTFINAL: the optimal solution for event location EVTSACF: event location stored in SAC file STATION: station sketch figure WAVE: waveform profile figure PKTCMP: comparison chart for current and previous picks

Table 3. Actions and operations in waveform profile

Actions	Operations
Double-click left mouse button	Shown manual picks as ticks or curve
Press & drag left mouse button	Select a time window, and then show its details in zoomed figure workspace (Figure 4). Note that, it is not workable if time window is < 10 screen units.
Click middle mouse button	Turn on/off theoretical predictions
Click middle mouse button on waveform curve	Select/deselect a waveform and highlight its ancillary information
Click right mouse button	Move the zoomed figure workspace to the nearest epicentral distance.
Move mouse cursor on prediction tick	Show phase name of prediction
Press key-x	Set up beginning/endding of the time window showing in zoomed figure workspace (same to press & drag left mouse button). Note that, it is not workable if time window is < 10 screen units.
Press key-space	Show/hide waveform

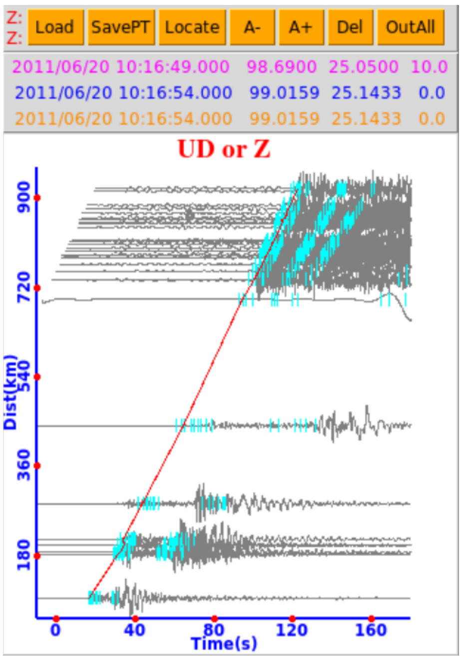


Figure 3. Waveform profile panel

Picking phase onset time by handwork

By pressing and dragging left mouse button in waveform profile, user can select a time window and inspect details of waveform in zoomed figure workspace (Figure 4). Note that, it is not workable if time window is < 10 screen units. Select/deselect checkboxes, ☐waveform ☐Envelope ☐Instfreq, seismic waveform, envelope, and instantaneous frequency will be shown/hidden, respectively. All of these curves are shown after normalization. The normalized height is set at Trace H: 100. Three values are embedded in button Trace H:. User is allowed to apply their preferred normalized height, which can be achieved by modifying the value in textbox adjacent to Trace H:. Please don't forget to click right mouse button on this textbox, in order to make it to be available. Two buttons, A- and A+, are used to scale up and scale down waveform amplitude, respectively.

User can enlarge workspace by clicking Enlarge measuring workspace button, or return back by clicking Shrink measuring workspace button. A time scale is plotted in top right corner of Figure 4. User can measure phase onset time through several actions as listed in Table 4.

A brief information (station ID, distance, azimuth, and number) is marked as a label tag to the left side of each waveform. The brief information is shown in two lines in label tag. First line is in order for station ID, distance and azimuth to epicenter, while second line is for the serial number. The font color of label tag would be red if a time tick is marked on its waveform (Figure 4), otherwise it is green. In some cases, the phase onset point may be not clear, which amplitude is suppressed by an inappropriate normalization value, such as those in top panel figure in Figure 5. It can be improved by pressing key-space on this workspace, which will re-normalize waveform amplitude in the current view window. Then the phase onset point shows up, as what are in bottom panel figure in Figure 5. Press key-space again, it will return.

Table 4. Actions and operations in zoomed figure workspace

Actions	Operations
Press & drag left mouse button	Zoom in waveform
Click right mouse button	Zoom out waveform
Double-click left mouse button	Mark a time tick on mouse cursor position
Press & drag left mouse button on time tick	Move time tick
Click middle mouse button on time tick	Delete time tick
Click middle mouse button	Turn on/off theoretical predictions
Click middle mouse button on waveform	Select/deselect a waveform and highlight its ancillary information
Click middle mouse button on station name label tag	Select/deselect a waveform and highlight its ancillary information
Move mouse cursor on prediction tick	Show phase name of prediction
Press key-x	Disable/enable green lines following mouse cursor
Press key-space	Re-normalize waveform amplitude in current view. See example in Figure 5

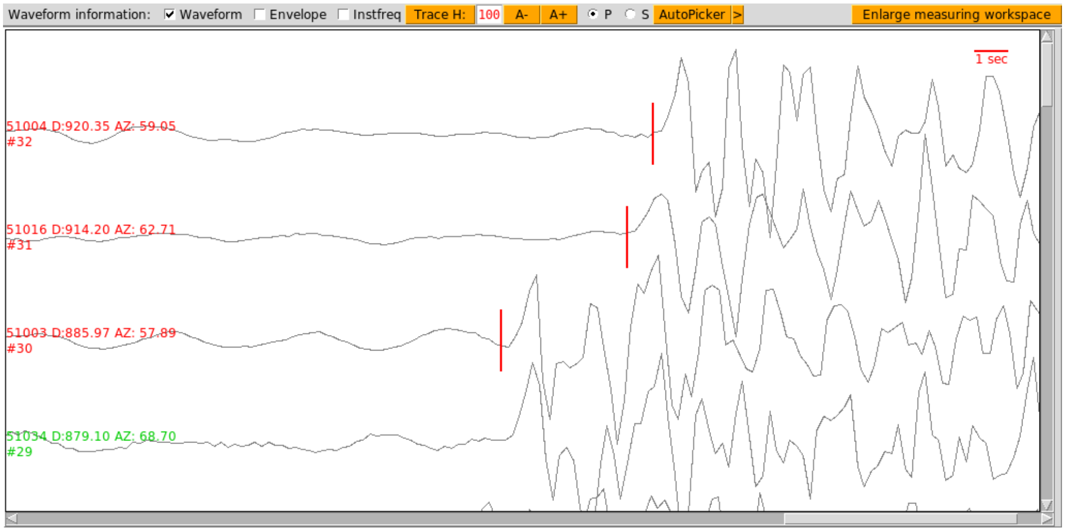


Figure 4. Zoomed figure workspace panel

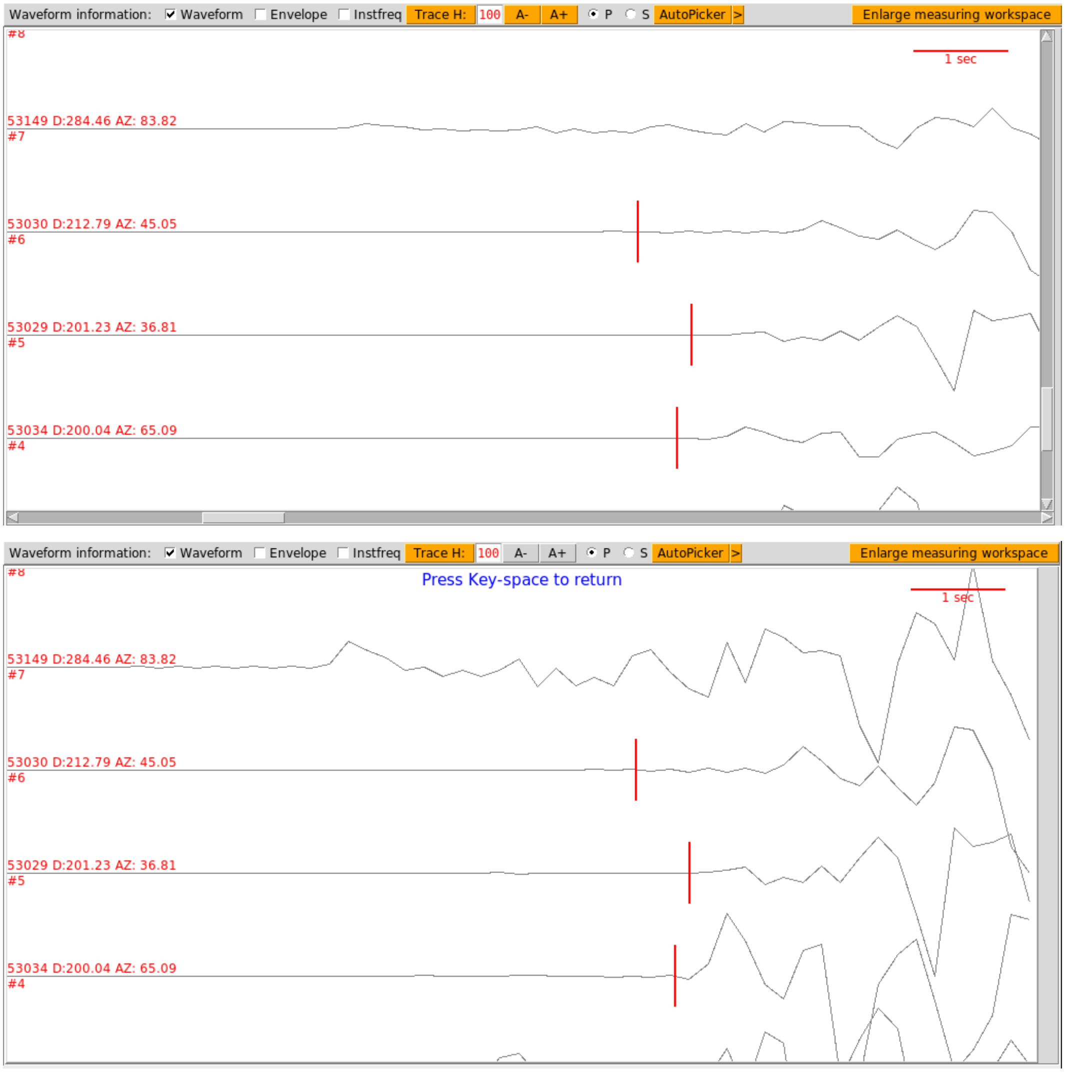
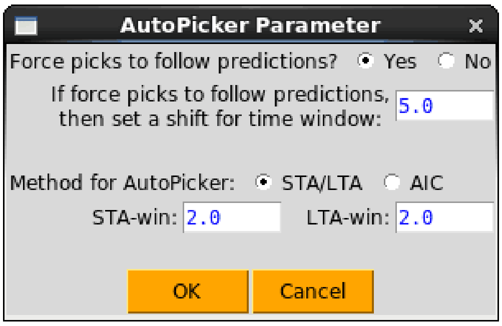


Figure 5. Example to show function of pressing key-space in zoomed figure workspace Top panel: waveform before pressing key-space; bottom panel: waveform after pressing key-space. The obvious difference is that waveform amplitude was re-normalized.

Picking phase onset time by automatic method

An AutoPicker is equipped in measurement procedure. The AutoPicker button is shown on the top of workspace. Two automatic algorithms are involved in AutoPicker, they are STA/LTA (Allen, 1978, 1982) and AIC (Akaike, 1974) algorithm. To set time window configuration is to click >, a pop up window will show as following:



AutoPicker are workable for both P- and S-wave phase. User can switch to the desired phase through two radio buttons, ⊙P ⊙S. After application of AutoPicker, user can modify those picks which are probably marked with errors.

Relocating hypocentral position

User can relocate hypocentral position by clicking Locate button. Hypocentral position is solved out through a grid-based search algorithm. Several groups of input arguments should be provided to this processing.

The first group of parameters is regional range in which the relocation procedure will search for hypocentral position. This group of input arguments is given in the format as following:

lonmin	*lonstep*	*lonmax*	*latmin*	*latstep*	*latmax*	*depmin*	*depstep*	*depmax*

Regional boundary: *lonmin, lonmax, latmin, latmax, depmin, depmax*

Grid interval: *lonstep, latstep, depstep*

Gray-shaded arguments (*lonmin, lonmax, latmin* and *latmax*) are automatically set by loading values from station sketch boundary (Figure 2). These four arguments are not fixed, user can modify them through changing station sketch view scope. White-shaded arguments (*lonstep, latstep, depmin, depstep* and *depmax*) are manually set by user. User can enter interval value through several built-in combinations in Locate para: button,.

The second group of parameters is velocity model. Optional velocity models are ak135 and iasp91, which can be switched by a group of radio buttons ⊙ak135 ⊙iasp91.

The third group of parameters is manual picks and error limitation. Manual picks are transmitted to relocation procedure via a file that is stored in output folder with file extension as PICK.FORLOC. Error limitation is a value defining the maximal deviation used by relocation procedure. The larger this value is, the more picks would be used to relocate hypocentral position. Error limitation is set at ErrLimit.

After relocation procedure, waveform profile will be automatically updated and two blue crosses (+) accompanied by location uncertainty (gray shaded circle in Figure 2) will appear in station sketch figure. Sometimes, these two blue crosses would overlap due to same results although the source positions are solved out by two algorithms. Relocation procedure will renew event position and accordingly update the associated waveform profile as event waveforms are sorted by epicentral distance.

Checking manual picks

TIMCPOT provides two ways to preliminarily evaluate manual picks. One way is to check consistency between picks which are, respectively, currently measured and previously identified in same area. The other way is to check the smoothness of traveltime curve. Two figure panels (Figure 6) will pop out for these two checking procedures which are called by clicking buttons check picks and check curve, respectively.

Both figures will be updated immediately once user marks or deletes a time tick. In left figure, the magenta circles are for those picks which are currently measured. The black dots are for those picks which were identified previously. Mouse buttons are bound with magenta circles to highlight ancillary information on manual picks. User can choose a radius value to define a circular area centered around the current event position. In this case, only those picks which are spatially close to the current event are shown, for the sake of reducing systematic errors. In right figure, red line segments and black dots compose a sketch for traveltime curve. Details of traveltime curve sketch can be inspected through mouse wheel and buttons. Table 5 lists available actions and its operations in both figure panels.

Table 5. Actions and operations

Actions	Operations
Double-click left mouse button on magenta circle (In left figure)	Select/deselect its associated waveform and highlight its ancillary information
Scroll mouse wheel (In right figure)	Zoom in/out traveltime curve
Press & drag left mouse button (In right figure)	Move traveltime curve
Double-click left mouse button on black dot (In right figure)	Select/deselect its associated waveform and highlight its ancillary information

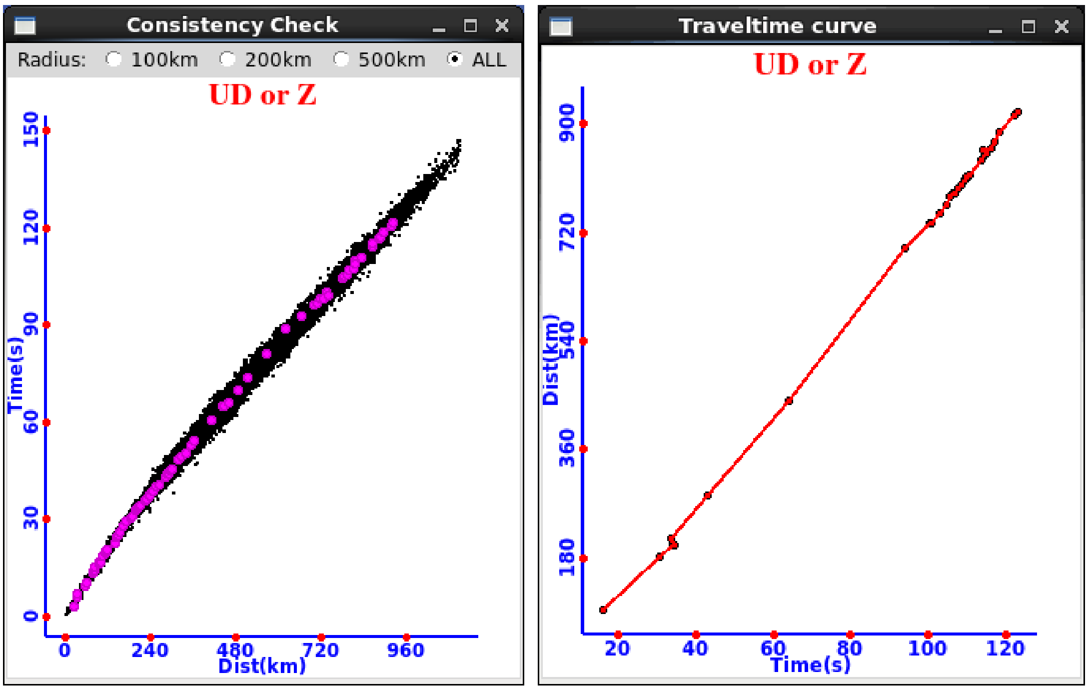


Figure 6. Two panels for comparison chart and traveltime curve

Exporting files and its format

TIMCPOT exports results into output folder, the exporting procedures are primarily triggered by three buttons:

By clicking SavePT button, TIMCPOT exports brief information on manual picks, just for relocation procedure.

By clicking Locate button, TIMCPOT exports relocation results.

By clicking OutAll button, TIMCPOT exports complete information on manual picks, event locations, and figures.

The file list is as following:

PKTRLOC: manually picking traveltime, brief information used in relocation procedure

PKTFINAL: manually picking traveltime, complete meta information for tomographic study

EVTRLOC: event location derived from relocate procedure

EVTFINAL: the optimal solution for event location

EVTSACF: event location stored in SAC file

STATION: station sketch figure

WAVE: waveform profile figure

PKTCMP: comparison chart for current and previous picks

There is an indicator (Figure 7) displaying if these files were exported or not. ZZ is for up-down component. ET is for east-west component. NR is for north-south component. If two horizontal channels of seismic data were rotated, then ET is for tangential component while NR is for radial component.

Manual picks and event locations are saved as plain text files which can be opened in any text editor. Figures are saved as postscript figure files, if command 'ps2raster' is available in user's Linux system, a copy of figures will be converted to JPG format files.

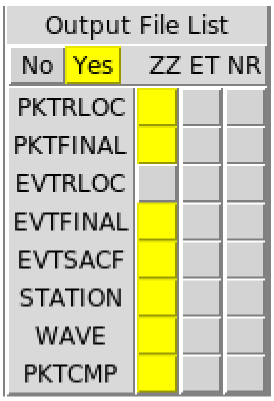


Figure 7. Indicator for file list

Manual picks are written out in plain text, and its format is as following:

```
  51030   27.8000  105.6000  1.2580 1   102.732 P   567.666 2011 171 10 16 49 000  5.500 2011 171 10 16 54 500    25.2000   98.9000   3.0  5.3  20110620.171.10.16.49.000_5.3  0.2000   6.5372  248.1840  65.2070
  53044   25.1000  100.5000  1.4020 1    31.434 P   103.468 2011 171 10 16 49 000  5.500 2011 171 10 16 54 500    25.2000   98.9000   3.0  5.3  20110620.171.10.16.49.000_5.3  0.2000   1.4535  274.2660  93.5897
  51027   28.0000  106.1000  1.1230 1   109.189 P   364.689 2011 171 10 16 49 000  5.500 2011 171 10 16 54 500    25.2000   98.9000   3.0  5.3  20110620.171.10.16.49.000_5.3  0.2000   7.0221  248.2421  65.0311
  51016   28.6000  107.0000  0.5610 1   121.994 P    83.450 2011 171 10 16 49 000  5.500 2011 171 10 16 54 500    25.2000   98.9000   3.0  5.3  20110620.171.10.16.49.000_5.3  0.2000   7.9841  246.7780  63.1263
  51025   28.0000  105.3000  0.7170 1   100.654 P   968.330 2011 171 10 16 49 000  5.500 2011 171 10 16 54 500    25.2000   98.9000   3.0  5.3  20110620.171.10.16.49.000_5.3  0.2000   6.3714  245.5018  62.6483
  53034   25.8000  100.5000  1.5420 1    34.262 P  8284.633 2011 171 10 16 49 000  5.500 2011 171 10 16 54 500    25.2000   98.9000   3.0  5.3  20110620.171.10.16.49.000_5.3  0.2000   1.5645  247.8934  67.2083
  51009   28.9000  106.1000  0.3430 1   114.194 P    43.298 2011 171 10 16 49 000  5.500 2011 171 10 16 54 500    25.2000   98.9000   3.0  5.3  20110620.171.10.16.49.000_5.3  0.2000   7.4012  241.8146  58.5525
  51026   28.0000  105.8000  0.6520 1   105.418 P   295.147 2011 171 10 16 49 000  5.500 2011 171 10 16 54 500    25.2000   98.9000   3.0  5.3  20110620.171.10.16.49.000_5.3  0.2000   6.7765  247.2721  64.1952
  51018   28.2000  105.2000  0.4670 1   100.337 P   424.148 2011 171 10 16 49 000  5.500 2011 171 10 16 54 500    25.2000   98.9000   3.0  5.3  20110620.171.10.16.49.000_5.3  0.2000   6.3774  243.5046  60.6858
  51033   27.7000  106.6000  1.0810 1   114.307 P   256.909 2011 171 10 16 49 000  5.500 2011 171 10 16 54 500    25.2000   98.9000   3.0  5.3  20110620.171.10.16.49.000_5.3  0.2000   7.3366  251.9034  68.4872
  53033   25.6000   99.4000  1.7050 1    16.256 P   356.023 2011 171 10 16 49 000  5.500 2011 171 10 16 54 500    25.2000   98.9000   3.0  5.3  20110620.171.10.16.49.000_5.3  0.2000   0.6026  228.7346  48.5213
  53030   26.4000  100.2000  2.0920 1    33.879 P   427.357 2011 171 10 16 49 000  5.500 2011 171 10 16 54 500    25.2000   98.9000   3.0  5.3  20110620.171.10.16.49.000_5.3  0.2000   1.6737  224.7241  44.1613
  51038   27.4000  107.0000  0.8270 1   116.815 P   996.437 2011 171 10 16 49 000  5.500 2011 171 10 16 54 500    25.2000   98.9000   3.0  5.3  20110620.171.10.16.49.000_5.3  0.2000   7.5921  255.0455  71.4705
  51032   27.7000  106.3000  0.8880 1   109.644 P  2923.549 2011 171 10 16 49 000  5.500 2011 171 10 16 54 500    25.2000   98.9000   3.0  5.3  20110620.171.10.16.49.000_5.3  0.2000   7.0843  251.0976  67.8148
  51004   29.1000  106.8000  0.4210 1   123.119 P   527.050 2011 171 10 16 49 000  5.500 2011 171 10 16 54 500    25.2000   98.9000   3.0  5.3  20110620.171.10.16.49.000_5.3  0.2000   8.0370  242.9468  59.3544
  51034   27.7000  107.0000  0.8940 1   117.747 P   151.005 2011 171 10 16 49 000  5.500 2011 171 10 16 54 500    25.2000   98.9000   3.0  5.3  20110620.171.10.16.49.000_5.3  0.2000   7.6747  252.9022  69.3081
  51019   28.3000  105.5000  0.3490 1   104.602 P   201.411 2011 171 10 16 49 000  5.500 2011 171 10 16 54 500    25.2000   98.9000   3.0  5.3  20110620.171.10.16.49.000_5.3  0.2000   6.6588  243.8906  60.9321
> 20110620.171.10.16.49.000_5.3
```

Explanatory notes to this plain text file are as following:

column #1: station ID, also known as STAID, station name and *kstnm*.
column #2-4: *latitude, longitude* and *elevation* for station.
column #5-6: switch (0/1), traveltime of manual picks.
    If switch is equal to 0, the manual pick is not available or seismic waveform signal noise ratio is too low to identify a clear phase onset time.
    if switch is equal to 1, an available manual pick is identified.
column #7: phase type. P-wave; S-wave; Undefined.
column #8: signal-to-noise ratio. It is calculate on a ratio of signal window and noise window.
column #9-14: reference time point for manual picks, listing in order as *year, Julian day*, *hour*, *minute, second, millisecond.*
column #15: time shift of event, refer to the time point appointed in column #9-14.
column #16-21: a time point when event occurred. An analogic equation for time is
    (column #9-14) + column #15 = (column #16-21)
column #22-25: *latitude, longitude*, *depth* and *magnitude* for event.
column #26: event ID, also known as EVTID.
column #27: data sample of seismic data from which user measured phase onset time.
column #28-30: distance, back-azimuth, and azimuth between station and event.

The last line is just a repetition of EVTID, but it begins with a char “>”.

The above explanatory notes are for complete meta information on manual picks saved in PTKFINAL. PKTRLOC contains brief information on manual picks, thus column #1-14.

Event locations are written out in plain text, and its format is as following:

```
20110620.171.10.16.49.000_5.3 2011/06/20 10:16:54.700    98.9244    25.1663   4.000    98.9244    25.1663   4.000  0.000000   0.00    2.0872    2.0874    0.1812    2.2891    0.0000    1.1662
```

Explanatory notes to this plain text file are as following:

column #1: event ID, also known as EVTID.
column #2-3: date and time for event.
column #4-6: position1 (longitude1, latitude1, and depth1) determined from minimal SD.
column #7-9: position2 (longitude2, latitude2, and depth2) determined from minimal RMS.
column #10: distance between position1 and position2.
column #11: difference between depth1 and depth2, that is depth1 minus depth2.
column #12-13: SD and RMS value of traveltime residuals.
column #14-15: Horizontal and depth uncertainty for hypocentral position determined by SD
column #16-17: Horizontal and depth uncertainty for hypocentral position determined by RMS

The above explanatory notes are for event location saved in EVTRLOC.
EVTFINAL and EVTSACF contain the first six columns, thus column #1-6.

Known problems

In current version (Version R1.0), most subroutines have been optimized to run with multiple threads. The drawing subroutine cannot be parallelly processed due to technical architecture of programing language. We have a plan to improve it in future development.

User's feedback

If user has any questions or problems when using TIMCPOT, please feel free to let me know. If user provides bug report, please attach test data and several specific operations which could trigger the bug. Innovated ideas to improve TIMCPOT are also welcome. The updated version is accessed by sending email to me. Not limitation to the issues mentioned above, anything about TIMCPOT, user can contact me via email (zhangfengxue336@163.com).